

Novelty Analysis

Analysis of Novelty, Transience, and Resonance

PENPAL Project

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Contents

0.1	Summary	1
1	1. Data Loading and Preprocessing	1
1.1	1.1 Load Data	1
1.2	1.2 Reshape Data	2
1.3	1.3 Descriptive Statistics	2
2	2. Novelty and Resonance Analysis	4
2.1	2.1 Resonance vs Novelty Plot	4
2.2	2.2 Transience vs Novelty Plot	5
3	3. Statistical Models	6
3.1	3.1 Resonance Model	6
3.2	3.2 Transience Model	7
4	4. Supplementary: Resonance vs Novelty by LLM Type	8
5	5. Session Information	10

0.1 Summary

- Humans produce higher novelty (surprisal) and higher transience than the model, indicating stronger deviations from earlier story context.
- Both agents display a positive novelty–resonance relationship (innovation bias), but the slope is steeper for humans.
- This indicates that surprising human turns exert greater forward influence on narrative development.
- AI contributions show lower variance and lower informational impact relative to human inputs.
- Humans are the primary drivers of narrative innovation and thematic redirection.

1 1. Data Loading and Preprocessing

1.1 1.1 Load Data

```
# Load Field Data - using standardized column names
df_res_field <- read_csv(here("data/human-ai/processed/novelty_scores.csv")) %>%
  rename(
    user = author_1,
    ai = author_2,
```

```

    user_surprise = author_1_surprise,
    ai_surprise = author_2_surprise,
    user_transience = author_1_transience,
    ai_transience = author_2_transience
  ) %>%
  filter(!is.na(user_surprise) & !is.na(ai_surprise)) %>%
  mutate(
    user_resonance = user_surprise - user_transience,
    ai_resonance = ai_surprise - ai_transience,
    datatype = "Field"
  ) %>%
  drop_na(user, ai) %>%
  filter(conversation_id != "conv_0bb56093-3033-4615-bb70-ebfa4135589a", conversation_id != "conv_0f18b")

```

1.2 1.2 Reshape Data

```

# Compute word counts for each agent's turn
df_res_field <- df_res_field %>%
  mutate(
    user_nwords = str_count(user, "\\S+"),
    ai_nwords = str_count(ai, "\\S+")
  )

df_res_long <- df_res_field %>%
  pivot_longer(
    cols = c(user_surprise, user_transience, user_nwords,
              ai_surprise, ai_transience, ai_nwords),
    names_to = c("agent", "metric"),
    names_pattern = "(user|ai)_(surprise|transience|nwords)"
  ) %>%
  pivot_wider(names_from = metric, values_from = value) %>%
  mutate(
    resonance = surprise - transience,
    agent = factor(agent, levels = c("user", "ai"), labels = c("User", "LLM"))
  )

```

1.3 1.3 Descriptive Statistics

```

desc_stats <- df_res_long %>%
  group_by(agent) %>%
  summarise(
    n = n(),
    mean_surprise = mean(surprise, na.rm = TRUE),
    sd_surprise = sd(surprise, na.rm = TRUE),
    mean_transience = mean(transience, na.rm = TRUE),
    sd_transience = sd(transience, na.rm = TRUE),
    mean_resonance = mean(resonance, na.rm = TRUE),
    sd_resonance = sd(resonance, na.rm = TRUE)
  )

print(desc_stats)

```

```
## # A tibble: 2 x 8
```

```
## agent      n mean_surprise sd_surprise mean_transience sd_transience
## <fct> <int>      <dbl>      <dbl>          <dbl>          <dbl>
## 1 User      767        -1.82        0.701          -1.02          0.447
## 2 LLM       767        -2.64        0.681          -0.566         0.432
## # i 2 more variables: mean_resonance <dbl>, sd_resonance <dbl>
knitr::kable(desc_stats, caption = "Descriptive Statistics for Novelty Metrics by Agent", digits = 3)
```

Table 1: Descriptive Statistics for Novelty Metrics by Agent

agent	n	mean_surprise	sd_surprise	mean_transience	sd_transience	mean_resonance	sd_resonance
User	767	-1.818	0.701	-1.022	0.447	-0.796	0.875
LLM	767	-2.640	0.681	-0.566	0.432	-2.048	0.807

```
agent_colors <- c("User" = "#377EB8", "LLM" = "#FF7F00")

df_metric_distributions <- df_res_long %>%
  select(agent, surprise, transience, resonance) %>%
  pivot_longer(
    cols = c(surprise, transience, resonance),
    names_to = "metric",
    values_to = "value"
  ) %>%
  mutate(
    metric = factor(
      metric,
      levels = c("surprise", "transience", "resonance"),
      labels = c("Novelty", "Transience", "Resonance")
    )
  )

metric_summary <- df_metric_distributions %>%
  group_by(metric, agent) %>%
  summarise(
    mean_value = mean(value, na.rm = TRUE),
    .groups = "drop"
  )

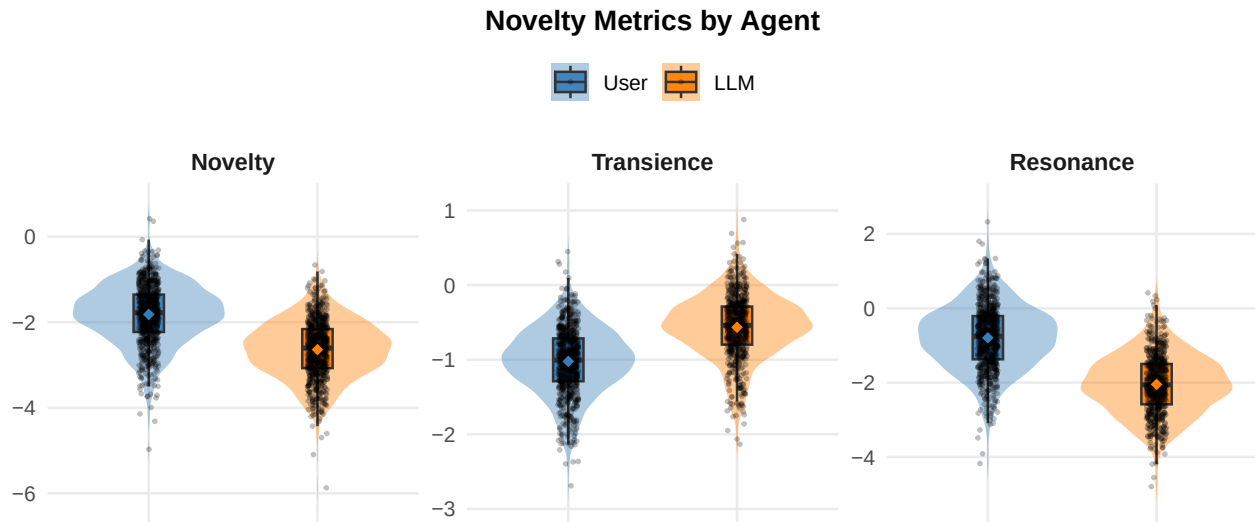
p_metric_distributions <- ggplot(
  df_metric_distributions,
  aes(x = agent, y = value, fill = agent)
) +
  geom_violin(alpha = 0.4, colour = NA, trim = FALSE, na.rm = TRUE) +
  geom_boxplot(width = 0.18, outlier.shape = NA, alpha = 0.9, na.rm = TRUE) +
  geom_jitter(width = 0.06, alpha = 0.25, size = 0.45, na.rm = TRUE) +
  geom_point(
    data = metric_summary,
    aes(x = agent, y = mean_value, color = agent),
    shape = 18,
    size = 2,
    show.legend = FALSE
  ) +
  facet_wrap(~ metric, nrow = 1, scales = "free_y") +
```

```

scale_fill_manual(values = agent_colors, name = NULL) +
scale_color_manual(values = agent_colors, name = NULL) +
labs(
  title = "Novelty Metrics by Agent",
  x = NULL,
  y = NULL
) +
theme_minimal(base_size = 11) +
theme(
  legend.position = "top",
  panel.grid.minor = element_blank(),
  strip.text = element_text(face = "bold", size = 10),
  plot.title = element_text(hjust = 0.5, size = 12, face = "bold"),
  axis.text.x = element_blank(),
  axis.ticks.x = element_blank(),
  plot.margin = margin(4, 4, 4, 4)
)

print(p_metric_distributions)

```



```

ggsave(
  file.path(GRAPHS_DIR, "Novelty_Transience_Resonance_Distributions_Agent.png"),
  p_metric_distributions,
  width = 7.5,
  height = 3.2,
  dpi = 300
)

```

2. Novelty and Resonance Analysis

2.1 Resonance vs Novelty Plot

```

# Plot Resonance vs Novelty (Field Data)
p_resonance <- ggplot(df_res_long, aes(x = surprise, y = resonance, color = agent)) +
  geom_point(alpha = 0.5, size = 1.5) +
  geom_smooth(method = "lm", se = TRUE, fullrange = TRUE, size = 1.2) +

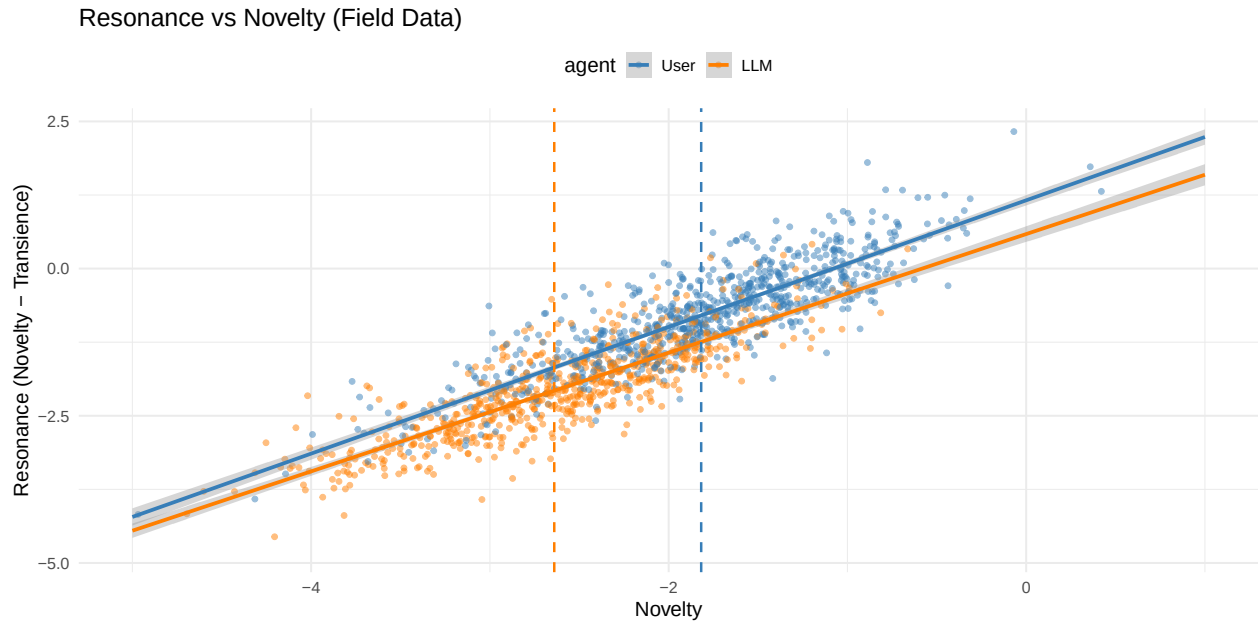
```

```

scale_color_manual(values = c("User" = "#377EB8", "LLM" = "#FF7F00")) +
  geom_vline(data = desc_stats,
    aes(xintercept = mean_surprise, color = agent),
    linetype = "dashed", linewidth = 0.8, show.legend = FALSE) +
  labs(
    title = "Resonance vs Novelty (Field Data)",
    x = "Novelty",
    y = "Resonance (Novelty - Transience)"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "top") + xlim(-5, 1)

print(p_resonance)

```



```

ggsave(file.path(GRAPHS_DIR, "novelty_resonance_vs_novelty.png"),
  p_resonance, width = 12, height = 6, dpi = 300)

```

2.2 Transience vs Novelty Plot

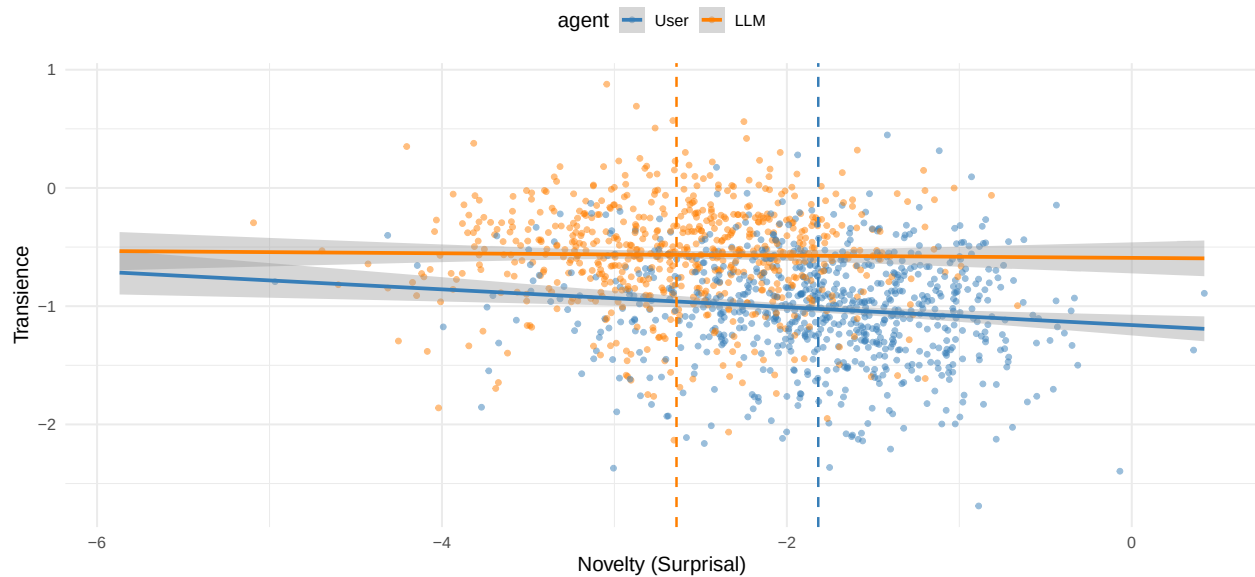
```

p_transience <- ggplot(df_res_long, aes(x = surprise, y = transience, color = agent)) +
  geom_point(alpha = 0.5, size = 1.5) +
  geom_smooth(method = "lm", se = TRUE, fullrange = TRUE, size = 1.2) +
  scale_color_manual(values = c("User" = "#377EB8", "LLM" = "#FF7F00")) +
  geom_vline(data = desc_stats,
    aes(xintercept = mean_surprise, color = agent),
    linetype = "dashed", linewidth = 0.8, show.legend = FALSE) +
  labs(
    title = "Transience vs Novelty (Field Data)",
    x = "Novelty (Surprisal)",
    y = "Transience"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "top")

```

```
print(p_transience)
```

Transience vs Novelty (Field Data)



```
ggsave(file.path(GRAPHS_DIR, "novelty_transience_vs_novelty.png"),
        p_transience, width = 12, height = 6, dpi = 300)
```

3 3. Statistical Models

3.1 3.1 Resonance Model

```
# Model without length control
model_resonance <- lmer(resonance ~ surprise * agent + (1 | conversation_id), data = df_res_long)
cat("=== Resonance Model ===\n")
```

```
## === Resonance Model ===
```

```
print(summary(model_resonance)$coefficients)
```

```
##              Estimate Std. Error      df    t value      Pr(>|t|)
## (Intercept)   1.21843066 0.04554494 1135.317  26.752275 1.180443e-122
## surprise      1.10789626 0.02242379 1429.730  49.407190 1.627902e-311
## agentLLM     -0.51501099 0.07999238 1427.903  -6.438250 1.646644e-10
## surprise:agentLLM -0.05528472 0.03345474 1428.208  -1.652523 9.864780e-02
```

```
cat("\n=== Slope Comparison (Resonance) ===\n")
```

```
##
```

```
## === Slope Comparison (Resonance) ===
```

```
print(emtrends(model_resonance, specs = "agent", var = "surprise"))
```

```
## agent surprise.trend      SE    df lower.CL upper.CL
## User              1.11 0.0224 1430      1.06      1.15
## LLM               1.05 0.0252 1434      1.00      1.10
##
```

```
## Degrees-of-freedom method: kenward-roger
```

```
## Confidence level used: 0.95
# Model WITH length control
model_resonance_len <- lmer(resonance ~ surprise * agent + nwords + (1 | conversation_id), data = df_res)
cat("\n=== Resonance Model (controlling for turn length) ===\n")

##
## === Resonance Model (controlling for turn length) ===
print(summary(model_resonance_len)$coefficients)

##              Estimate Std. Error      df    t value      Pr(>|t|)
## (Intercept)    1.129046654 0.080002370  990.5451 14.112665 2.410673e-41
## surprise       1.104138524 0.022744560 1423.1627 48.545170 3.672747e-304
## agentLLM      -0.536142108 0.081728249 1419.5108 -6.560059 7.515289e-11
## nwords         0.003219824 0.002282028  836.2298  1.410948 1.586318e-01
## surprise:agentLLM -0.058077118 0.033510812 1420.8084 -1.733086 8.329736e-02
cat("\n=== Slope Comparison (Resonance, length-controlled) ===\n")

##
## === Slope Comparison (Resonance, length-controlled) ===
print(emtrends(model_resonance_len, specs = "agent", var = "surprise"))

## agent surprise.trend      SE    df lower.CL upper.CL
## User              1.10 0.0228 1425     1.059     1.15
## LLM               1.05 0.0260 1428     0.995     1.10
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

3.2 Transience Model

```
# Model without length control
model_transience <- lmer(transience ~ surprise * agent + (1 | conversation_id), data = df_res_long)
cat("=== Transience Model ===\n")

## === Transience Model ===
print(summary(model_transience)$coefficients)

##              Estimate Std. Error      df    t value      Pr(>|t|)
## (Intercept)   -1.21843066 0.04554494 1135.317 -26.752275 1.180444e-122
## surprise      -0.10789626 0.02242379 1429.730 -4.811688 1.655055e-06
## agentLLM       0.51501099 0.07999238 1427.903  6.438250 1.646644e-10
## surprise:agentLLM 0.05528472 0.03345474 1428.208  1.652523 9.864780e-02
cat("\n=== Slope Comparison (Transience) ===\n")

##
## === Slope Comparison (Transience) ===
print(emtrends(model_transience, specs = "agent", var = "surprise"))

## agent surprise.trend      SE    df lower.CL upper.CL
## User        -0.1079 0.0224 1430    -0.152 -0.06386
## LLM         -0.0526 0.0252 1434    -0.102 -0.00316
```

```
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
# Model WITH length control
model_transience_len <- lmer(transience ~ surprise * agent + nwords + (1 | conversation_id), data = df_)

cat("\n=== Transience Model (controlling for turn length) ===\n")

##
## === Transience Model (controlling for turn length) ===
print(summary(model_transience_len)$coefficients)

##              Estimate Std. Error      df    t value      Pr(>|t|)
## (Intercept)   -1.129046654 0.080002370  990.5451 -14.112665 2.410673e-41
## surprise      -0.104138524 0.022744560 1423.1627  -4.578612 5.089017e-06
## agentLLM       0.536142108 0.081728249 1419.5108   6.560059 7.515289e-11
## nwords        -0.003219824 0.002282028  836.2298  -1.410948 1.586318e-01
## surprise:agentLLM 0.058077118 0.033510812 1420.8084   1.733086 8.329736e-02

cat("\n=== Slope Comparison (Transience, length-controlled) ===\n")

##
## === Slope Comparison (Transience, length-controlled) ===
print(emtrends(model_transience_len, specs = "agent", var = "surprise"))

## agent surprise.trend      SE    df lower.CL upper.CL
## User          -0.1041 0.0228 1425   -0.149  -0.0595
## LLM           -0.0461 0.0260 1428   -0.097   0.0049
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

4 4. Supplementary: Resonance vs Novelty by LLM Type

```
desc_stats_llm <- df_res_long %>%
  filter(!is.na(llm_type)) %>%
  group_by(agent, llm_type) %>%
  summarise(mean_surprise = mean(surprise, na.rm = TRUE), .groups = "drop")

p_llm_user <- df_res_long %>%
  filter(!is.na(llm_type), agent == "User") %>%
  ggplot(aes(x = surprise, y = resonance, color = factor(llm_type))) +
  geom_point(alpha = 0.4, size = 1.2) +
  geom_smooth(method = "lm", se = TRUE, linewidth = 1.2) +
  geom_vline(data = desc_stats_llm %>% filter(agent == "User"),
    aes(xintercept = mean_surprise, color = factor(llm_type)),
    linetype = "dashed", linewidth = 0.8, show.legend = FALSE) +
  labs(
    title = "User: Resonance vs Novelty by LLM Type",
    x = "Novelty",
    y = "Resonance (Novelty - Transience)",
    color = "LLM Type"
  ) +
```



```

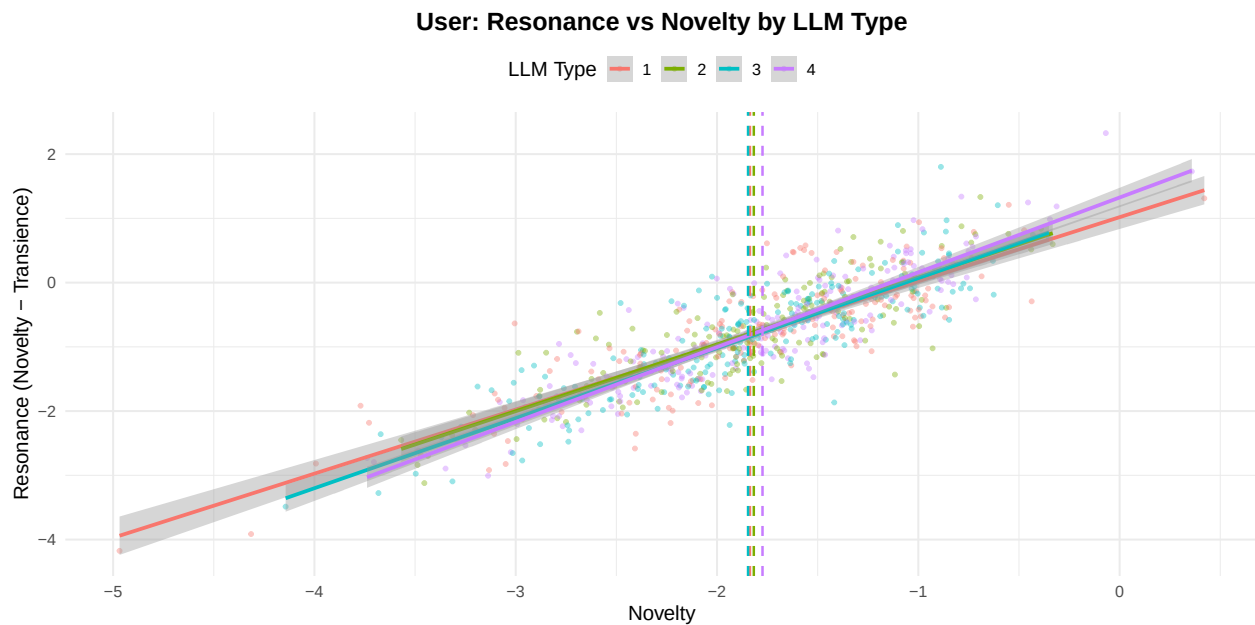
theme_minimal(base_size = 14) +
theme(legend.position = "top",
      plot.title = element_text(hjust = 0.5, face = "bold"))

p_llm_ai <- df_res_long %>%
  filter(!is.na(llm_type), agent == "LLM") %>%
  ggplot(aes(x = surprise, y = resonance, color = factor(llm_type))) +
  geom_point(alpha = 0.4, size = 1.2) +
  geom_smooth(method = "lm", se = TRUE, linewidth = 1.2) +
  geom_vline(data = desc_stats_llm %>% filter(agent == "LLM"),
            aes(xintercept = mean_surprise, color = factor(llm_type)),
            linetype = "dashed", linewidth = 0.8, show.legend = FALSE) +

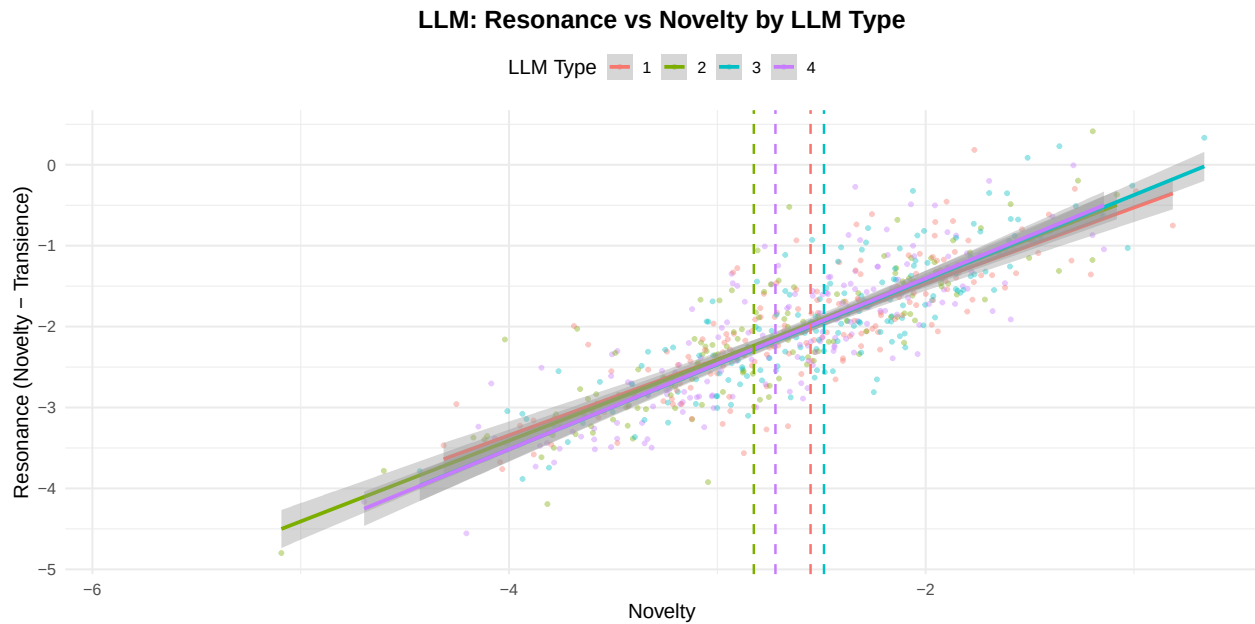
  labs(
    title = "LLM: Resonance vs Novelty by LLM Type",
    x = "Novelty",
    y = "Resonance (Novelty - Transience)",
    color = "LLM Type"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "top",
        plot.title = element_text(hjust = 0.5, face = "bold"))

print(p_llm_user)

```



```
print(p_llm_ai)
```



```
ggsave(file.path(GRAPHS_DIR, "novelty_resonance_llm_type_user.png"),
  p_llm_user, width = 12, height = 6, dpi = 300)
ggsave(file.path(GRAPHS_DIR, "novelty_resonance_llm_type_ai.png"),
  p_llm_ai, width = 12, height = 6, dpi = 300)
```

5 5. Session Information

```
sessionInfo()
```

```
## R version 4.4.2 (2024-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS 26.3
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8
##
## time zone: Australia/Melbourne
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices datasets  utils      methods    base
##
## other attached packages:
## [1] emmeans_2.0.2  here_1.0.2    afex_1.4-1    cowplot_1.1.3
## [5] patchwork_1.3.0 zoo_1.8-12     stringi_1.8.4 arrow_19.0.1.1
## [9] lmerTest_3.1-3 lme4_1.1-35.5 Matrix_1.7-1  lubridate_1.9.3
## [13] forcats_1.0.0 stringr_1.5.1 dplyr_1.1.4   purrr_1.2.1
```

```
## [17] readr_2.1.5      tidyr_1.3.1      tibble_3.2.1     ggplot2_3.5.1
## [21] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] tidyselect_1.2.1   farver_2.1.2      fastmap_1.2.0
## [4] pacman_0.5.1       digest_0.6.37     timechange_0.3.0
## [7] estimability_1.5.1 lifecycle_1.0.4    magrittr_2.0.3
## [10] compiler_4.4.2     rlang_1.1.7       tools_4.4.2
## [13] utf8_1.2.4         yaml_2.3.10       knitr_1.50
## [16] labeling_0.4.3     bit_4.0.5         plyr_1.8.9
## [19] abind_1.4-8        withr_3.0.2       numDeriv_2016.8-1.1
## [22] grid_4.4.2         fansi_1.0.6       colorspace_2.1-1
## [25] scales_1.3.0       MASS_7.3-61       tinytex_0.52
## [28] cli_3.6.3          mvtnorm_1.3-6     rmarkdown_2.28
## [31] crayon_1.5.3       ragg_1.3.2        generics_0.1.3
## [34] reshape2_1.4.4     tzdb_0.4.0        minqa_1.2.8
## [37] splines_4.4.2      assertthat_0.2.1  parallel_4.4.2
## [40] vctrs_0.7.2        boot_1.3-31       carData_3.0-5
## [43] car_3.1-3          hms_1.1.3         bit64_4.0.5
## [46] pbkrtest_0.5.3     Formula_1.2-5     systemfonts_1.1.0
## [49] glue_1.7.0         nloptr_2.1.1      gtable_0.3.5
## [52] munsell_0.5.1      pillar_1.9.0      htmltools_0.5.8.1
## [55] R6_2.5.1           textshaping_0.4.0 rprojroot_2.1.1
## [58] vroom_1.6.5        evaluate_1.0.3    lattice_0.22-6
## [61] backports_1.5.0    broom_1.0.6       renv_1.1.5
## [64] Rcpp_1.0.13        coda_0.19-4.1     nlme_3.1-166
## [67] mgcv_1.9-1         xfun_0.54         pkgconfig_2.0.3
```

Analysis complete. All figures saved to `analysis/figures/` and tables to `analysis/tables/`.